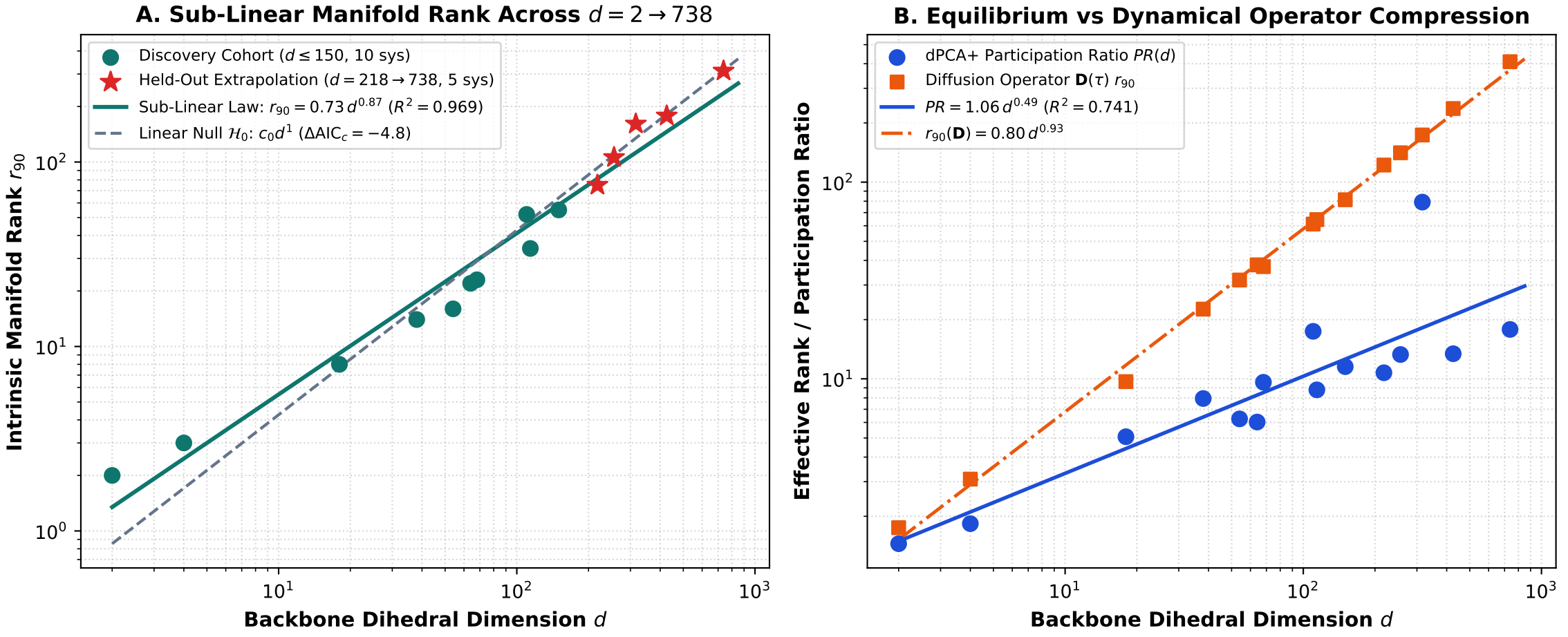
A. Sub-Linear Manifold Rank Across $d = 2 \rightarrow 738$
Discovery Cohort ($d \leq 150$, 10 sys)
Held-Out Extrapolation ($d = 218 \rightarrow 738$, 5 sys)
Sub-Linear Law: $r_{90} = 0.73\, d^{0.87}$ ($R^2 = 0.969$)
Linear Null $\mathcal{H}_0$: $c_0 d^1$ ($\Delta\text{AIC}_c = -4.8$)
Intrinsic Manifold Rank r_{90}
Backbone Dihedral Dimension d
B. Equilibrium vs Dynamical Operator Compression
dPCA+ Participation Ratio $PR(d)$
Diffusion Operator $\mathbf{D}(\tau)$ r_{90}
$PR = 1.06\, d^{0.49}$ ($R^2 = 0.741$)
$r_{90}(\mathbf{D}) = 0.80\, d^{0.93}$
Effective Rank / Participation Ratio
Backbone Dihedral Dimension d